

Digital Forensics

Introduction

- Digital Forensics is the process of identifying, collecting, preserving, analyzing, and presenting digital evidence from electronic devices.

Why Digital Forensics is Important

- • Investigate cybercrimes
- • Identify hackers
- • Recover deleted data
- • Provide legal evidence
- • Improve cybersecurity

Real World Example – Ransomware Attack

- A hospital system is locked by ransomware. Investigators analyze infected computers, track the attack source, and identify the phishing email used to deliver malware.

Digital Forensics Investigation Process

- 1. Identification
- 2. Preservation
- 3. Collection
- 4. Examination
- 5. Analysis
- 6. Reporting

1. Identification

Description

Identification is the first step where investigators determine **which devices or systems may contain digital evidence** related to the incident.

The goal is to locate **potential sources of evidence**.

Possible Evidence Sources

Computers

Mobile phones

Hard drives

USB drives

1. Identification

Example

A company suspects an employee of stealing confidential files.

Investigators identify:

The employee's laptop

USB drives used

Company server logs

Tools Used

Autopsy (digital forensics platform)

EnCase Forensic

2. Preservation

- **Description**
- **Preservation** ensures that digital evidence is **protected from modification, deletion, or damage**.
- Digital evidence can easily change, so investigators must **secure it immediately**.
- **Key Activities**
- Disconnect the system from the network
- Prevent unauthorized access
- Use write blockers
- Create forensic images

2. Preservation

- **Example**
- Instead of analyzing the original hard drive, investigators create a **forensic copy** so the original evidence remains unchanged.
- **Important Concept**
- **Chain of Custody**
- This documents:
- Who collected the evidence
- When it was collected
- Where it was stored
- This ensures evidence is **legally valid in court**.

3. Collection

- **Description**
- In this stage investigators **collect digital evidence from identified sources**.
- This must be done carefully to avoid altering data.
- **Data Collected**
- Disk images
- RAM memory dumps
- Log files
- Emails
- Network traffic
- **Example**
- Investigators copy the suspect's **hard drive and USB data** for analysis.

3. Collection

- **Tools Used**
- FTK (Forensic Toolkit)
- EnCase Forensic
- FTK Imager

4. Examination

- **Description**
- The **examination phase** involves searching and extracting relevant information from the collected evidence.
- Investigators filter out unnecessary data and locate useful evidence.
- **Activities**
- Recover deleted files
- Extract hidden data
- Analyze internet history
- Examine emails
- Identify suspicious programs

4. Examination

- **Example**
- An investigator recovers **deleted company documents** that the suspect tried to remove.
- **Tools Used**
- Autopsy (digital forensics platform)
- FTK (Forensic Toolkit)

5. Analysis

- **Description**
- During the **analysis phase**, investigators interpret the extracted data to understand **what happened, how it happened, and who was responsible**.
- This step reconstructs the **timeline of events**.
- **Key Questions**
- When did the attack occur?
- Who accessed the system?
- What files were accessed?
- How did the attacker gain access?

5. Analysis

Example Timeline data theft

Time	Activity
10:05 AM	USB drive inserted
10:07 AM	Confidential file copied
10:10 AM	USB removed

5. Analysis

- **Tools Used**
- Wireshark for network traffic analysis
- Volatility (memory forensics framework) for memory analysis

6. Reporting

- **Description**
- The final step is creating a **clear and detailed forensic report** that explains the findings.
- This report may be used in:
 - Legal cases
 - Corporate investigations
 - Cybersecurity incident reports

6. Reporting

- **Contents of the Report**
- Investigation objective
- Methods used
- Evidence collected
- Analysis results
- Timeline of events
- Final conclusion

6. Reporting

- **Example Conclusion**
- "The investigation shows that confidential company files were copied to a USB device on March 10 at 10:07 AM."
- **Importance**
- The report must be:
- Accurate
- Clear
- Easy to understand
- Legally acceptable

Summary of digital forensics Process

Step	Purpose
Identification	Locate potential evidence
Preservation	Protect evidence from modification
Collection	Gather digital data
Examination	Extract relevant information
Analysis	Understand what happened
Reporting	Document investigation findings

Real World Example

- **Incident**
- A company suspects an employee copied confidential data.

Digital Forensics Investigation

1 Identification

Identify employee computer and USB devices.

2 Preservation

Create forensic image of the computer.

3 Collection

Collect disk image and system logs.

4 Examination

Recover deleted documents.

5 Analysis

Find that files were copied to USB at 10:07 AM.

6 Reporting

Prepare report proving data theft.



Types of Digital Forensics

- • Computer Forensics
- • Network Forensics
- • Mobile Forensics
- • Email Forensics
- • Malware Forensics

Computer Forensics

- Focuses on analyzing computers and storage devices to recover files, investigate user activity, and detect malicious actions.

Network Forensics

- Analyzes network traffic to detect attacks, trace hacker IP addresses, and identify suspicious communication.

Mobile Forensics

- Investigates smartphones and tablets to recover messages, call logs, photos, and app data.

Challenges in Digital Forensics

- • Large amounts of data
- • Encryption
- • Cloud storage
- • Anti-forensics techniques used by attackers

Example Career Progression

Example timeline for a digital forensic professional:

Year	Job Role	Salary
0–1 years	Cybersecurity Intern	30k – 40k
1–3 years	Security Analyst / Junior Forensic Investigator	70k – 120k
3–5 years	Digital Forensic Investigator	150k – 250k
6–10 years	Senior Cyber Forensic Expert	300k – 400k+

Conclusion

- Digital Forensics plays a critical role in cybersecurity by helping organizations detect cyberattacks, investigate incidents, and protect digital systems.

Thank You

- Questions?